Type MS · Volume of revolution about the x-axis

Answer 1 of 6 · Question 6 part b · 6 marks

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6 a	$\cos 2A = \cos^2 A - \sin^2 A$ $= (1-\sin^2 A) - \sin^2 A = 1 - 2\sin^2 A$ *	M1 A1 cso (2)
b	$\pi\int_0^{\frac{\pi}{4}} (3+2\sin x)^2 \mathrm{d}x$	M1
	$\pi\int_0^{\frac{\pi}{4}} (9+12\sin x+4\sin^2 x)\,\mathrm{d}x$	M1
	$\pi\int_0^{\frac{\pi}{4}} (11+12\sin x-2\cos 2x)\,\mathrm{d}x$	M1
	$\pi[11x-12\cos x-\sin 2x]_0^{\frac{\pi}{4}}$	M1
	$\pi\left[\left(\frac{11\pi}{4}-6\sqrt{2}-1\right)-(0-12-0)\right]$ = awrt 35	M1 A1 (6)
		Total 8 marks

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Answer 1 of 6 · Question 6 part b · 6 marks

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Part	Mark	Notes
(a)	**M1**	For the use of both $\cos 2A = \cos^2 A - \sin^2 A$ and $\cos^2 A = \left(1 - \sin^2 A\right)$ to attempt to form the required identity.
	A1	For $\cos 2A = 1 - 2\sin^2 A$ with no errors seen.
(b)	**M1**	For the correct use of use of $\pi\int_a^b y^2 \mathrm{d}x$ to obtain: $\pi\int_0^{\frac{\pi}{4}} (3 + 2\sin x)^2 \mathrm{d}x$ with the correct limits and π You may not see π and the limits at this stage. They may add these in at the end of their work. Check and if that is the case, award this mark.
	M1	For expanding the given quadratic correctly to achieve: This must be correct. $\pi\int_0^{\frac{\pi}{4}} \left(9 + 12\sin x + 4\sin^2 x\right) \mathrm{d}x$ or accept $\pi\int_0^{\frac{\pi}{4}} \left(9 + 6\sin x + 6\sin x + 4\sin^2 x\right) \mathrm{d}x$ Ignore limits and π for this mark. **This is an A mark in Epen**
	M1	For using $\sin^2 A = \frac{1 - \cos 2A}{2}$ and substituting this identity into their expansion to attempt to obtain: $\pi\int_0^{\frac{\pi}{4}} (11 + 12\sin x - 2\cos 2x) \mathrm{d}x$ As a minimum we need to see: $A + B\sin x \pm C\cos 2x$ Ignore limits and π for this mark.
	M1	For an attempt to integrate their expression which must be of the form: $A + B\sin x \pm C\cos 2x$ **or** $A + B\sin x \pm C\sin^2 x$ where there must be an acceptable attempt to integrate as a minimum at least **both** A **and** $B\sin x$ to obtain $A \to Ax$ and $B\sin x \to -B\cos x$ Ignore limits and π for this mark.
	M1	For the correct substitution of the correct given limits the correct way around into an integrated/changed expression. $\pi\left[\left(\frac{11\pi}{4} - 6\sqrt{2} - 1\right) - (0 - 12 - 0)\right] = 35.041....$ This mark can be implied by the correct final answer **ONLY** if the correct integration is seen explicitly in the response. If their integrated expression is incorrect, and no substitution is seen award M0 here.
	A1	For awrt 35

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Question number	Scheme	Marks
	$\tan\theta + \tan\theta$	

Type MS · Volume of revolution about the x-axis

Answer 2 of 6 · Question 11 part b · 4 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

11 (a)	$\frac{e^{4x}}{32}$ **and** $8x^2-4x+1$	M1
	$\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{e^{4x}}{8}(8x^2-4x+1)$	A1A1
	$\frac{dy}{dx}=\frac{16xe^{4x}}{32}-\frac{4e^{4x}}{32}+\frac{8x^2e^{4x}}{8}-\frac{4xe^{4x}}{8}+\frac{e^{4x}}{8}$	dM1
	$\frac{dy}{dx}=x^2e^{4x}$ *	A1 cso [5]
(b)	Volume $=\pi\int_{-2}^{0}(3xe^{2x})^2\,dx$	M1
	$V=\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4x}}{32}(8x^2-4x+1)\right]_{-2}^{0}$	M1
	$\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4\times0}}{32}(8\times0^2-4\times0+1)\right]-\pi\left[\frac{9e^{4\times-2}}{32}(8\times-2^2-4\times-2+1)\right]$	dM1
	$V=0.87142.....\approx0.87$ (2sf)	A1 [4]
	SC - attempts integration by parts.	
	Volume $=\pi\int_{-2}^{0}(3xe^{2x})^2\,dx$	M1
	$\int x^2e^{4x}\,dx=\frac{x^2e^{4x}}{'4'}-\int2x\frac{e^{4x}}{'4'}\,dx$ $\int x^2e^{4x}\,dx=\frac{x^2e^{4x}}{'4'}-\left[\frac{2xe^{4x}}{'16'}-\int\frac{2e^{4x}}{'16'}\right]=\frac{x^2e^{4x}}{'4'}\pm\frac{2xe^{4x}}{'16'}\pm\frac{e^{4x}}{'32'}$	M1
	$\pi\int_{-2}^{0}9x^2e^{4x}\,dx=9\pi\left[\frac{x^2e^{4x}}{'4'}-\frac{2xe^{4x}}{'16'}+\frac{e^{4x}}{'32'}\right]_{-2}^{0}$ $=9\pi\left[\left(0-0+\frac{1}{32}\right)-\left(\frac{(-2)^2e^{4(-2)}}{'4'}-\frac{2(-2)e^{4(-2)}}{'16'}+\frac{e^{4(-2)}}{'32'}\right)\right]$	dM1
	For the correct volume of $0.87142.....\approx0.87$ rounded correctly to 2sf	A1 [4]
		Total 9 marks

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Answer 2 of 6 · Question 11 part b · 4 marks

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Part	Mark	Notes
(a)	**M1**	For using product rule correctly with an attempt to differentiate both $\frac{e^{4x}}{32}$ **and** $8x^2-4x+1$ Correct application of $\frac{dy}{dx}=u\frac{dv}{dx}+v\frac{du}{dx}$ is required For attempt to differentiate: $8x^2-4x+1\rightarrow 16x-4$ $\frac{e^{4x}}{32}\rightarrow ke^{4x}$ where $k\neq 1$ or $\frac{1}{32}$ or $\frac{1}{128}$
	A1	For either $\frac{e^{4x}}{32}(16x-4)$ OR $\frac{e^{4x}}{8}(8x^2-4x+1)$ o.e. in both cases
	A1	For $\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{e^{4x}}{8}(8x^2-4x+1)$ fully correct Accept $\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{4e^{4x}}{32}(8x^2-4x+1)$
	dM1	For multiplying out both sets of brackets, or factorising, their $\frac{dy}{dx}$ provided the earlier M mark has been achieved e.g., $\frac{dy}{dx}=\frac{16xe^{4x}}{32}-\frac{4e^{4x}}{32}+\frac{8x^2e^{4x}}{8}-\frac{4xe^{4x}}{8}+\frac{e^{4x}}{8}$ Some working to collect terms between product rule and final expression.
	A1 cso	For the correct expression only $\frac{dy}{dx}=x^2e^{4x}$ *
(b)	**M1**	This is a strategy mark for using the correct expression with the correct limits for the volume of rotation. Volume $=\pi\int_{-2}^{0}\left(3xe^{2x}\right)^2\,dx$
	M1	For using the **given** result from part (a) to integrate $V=\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4x}}{32}(8x^2-4x+1)\right]_{-2}^{0}$ Ignore π and limits for this mark even if they are missing or incorrect.
	dM1	For applying the correct limits in an attempt to evaluate the integral $\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4\times0}}{32}(8\times0^2-4\times0+1)\right]-\pi\left[\frac{9e^{4\times-2}}{32}(8\times-2^2-4\times-2+1)\right]$ Dependent on previous method mark. Condone omission of π for this mark.
	A1	For the correct volume of $0.87142\ldots\approx 0.87$ rounded correctly to 2sf
SC -attempts integration by parts.		
	M1	This is a strategy mark for using the correct expression with the correct limits for the volume of rotation.

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Type MS · Volume of revolution about the x-axis

Answer 2 of 6 · Question 11 part b · 4 marks

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		Volume $= \pi\int_{-2}^{0}\left(3xe^{2x}\right)^2\ dx$
	M1	For an attempt to integrate by parts. • They must use the correct formula • $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$ **TWICE** in the correct direction First integration – ignore π and 3^2 for this mark $\int x^2e^{4x}\,dx = \frac{x^2e^{4x}}{'4'} - \int 2x\frac{e^{4x}}{'4'}\,dx$ Second integration $\int x^2e^{4x}\,dx = \frac{x^2e^{4x}}{'4'} - \left[\frac{2xe^{4x}}{'16'} - \int\frac{2e^{4x}}{'16'}\right] = \frac{x^2e^{4x}}{'4'} \pm \frac{2xe^{4x}}{'16'} \pm \frac{e^{4x}}{'32'}$
	dM1	For applying the correct limits in an attempt to evaluate the integral $\pi\int_{-2}^{0}9x^2e^{4x}\,dx = 9\pi\left[\frac{x^2e^{4x}}{'4'} - \frac{2xe^{4x}}{'16'} + \frac{e^{4x}}{'32'}\right]_{-2}^{0}$ $= 9\pi\left[\left(0-0+\frac{1}{32}\right) - \left(\frac{(-2)^2e^{4(-2)}}{'4'} - \frac{2(-2)e^{4(-2)}}{'16'} + \frac{e^{4(-2)}}{'32'}\right)\right]$ Dependent on previous method mark. Condone omission of π for this mark.
	A1	For the correct volume of $0.87142..... \approx 0.87$ rounded correctly to 2sf

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Type MS · Volume of revolution about the x-axis

Answer 2 of 6 · Question 11 part b · 4 marks

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Type MS · Volume of revolution about the x-axis

Answer 3 of 6 · Question 11 part whole · 7 marks

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11	Intersection of the curve with the x-axis to find lower limit $\{y=\sqrt{4x-8}=0 \Rightarrow 4x-8=0\} \Rightarrow x=2$	B1
	Volume of rotation	
	$V=\pi\int_2^b\left(\sqrt{4x-8}\right)^2\,\mathrm{d}x=\left\{\pi\int_2^b(4x-8)\,\mathrm{d}x\right\}$	M1
	$V=\pi\left[\frac{4x^2}{2}-8x\right]_2^b$	M1
	$50\pi=\pi\left[\left(\frac{4\times b^2}{2}-8\times b\right)-\left(\frac{4\times 2^2}{2}-8\times 2\right)\right]$	M1
	$50=2b^2-8b+8 \Rightarrow 0=b^2-4b-21$ $0=(b+3)(b-7)$ $\Rightarrow b=7$	dddM1 ddddM1 A1 [7]
		Total 7 marks

Mark	Additional Guidance
B1	Finds the lower limit of the integral: For $x=2$ or (2,0) seen/embedded in their integral/on the graph
M1	Sets up a correct expression for the volume with their lower limit (a numerical value) found earlier and the upper limit of b, must have π
M1	Correctly integrates $4x-8$, with or without π
M1	Substitutes their lower limit (a numerical value) and upper limit b correctly into **their integrated** expression (with or without π), and equates to 50π / 50
dddM1	Forms a 3TQ in b/x Dependent on previous three M marks
ddddM1	Solves the 3TQ with any correct method Can be implied by their correct b value from a correct 3TQ Dependent on all previous M marks
A1	For $b=7$ only, from correct working, must be b=7, not x =7
Note: If attempted the volume of revolution about the y-axis, only B1 could be possibly awarded, no other marks could be scored.	

Type MS · Volume of revolution about the x-axis

Answer 3 of 6 · Question 11 part whole · 7 marks

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Type MS · Volume of revolution about the x-axis

Answer 4 of 6 · Question 2 part whole · 4 marks

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2	$\text{Vol} = \pi\int_0^3 \left(e^{3x}\right)^2 dx \left(= \pi\int_0^3 e^{6x} dx\right)$	M1
	$\pi\left[\frac{1}{6}e^{6x}\right]_0^3, = \left(\frac{1}{6}e^{18} - \frac{1}{6}\right)\pi$ oe	dM1A1,A1 (4) [4]

M1	Use $\text{Vol} = \pi\int y^2 dx$ Award if pi missing here but reappears later. Limits not needed, ignore any shown. dx may be missing.
dM1	Square correctly and attempt the integration. $e^{6x} \to ke^{6x}$ where $k = \pm\frac{1}{6}$ or ± 1 Limits and dx may be missing. Award if pi missing here but reappears later.
A1	Correct integration including correct limits
A1	Substitute the limits and obtain the correct answer

Type MS · Volume of revolution about the x-axis

Answer 4 of 6 · Question 2 part whole · 4 marks

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Question Number	Scheme	Marks
	$(1+px)^{-5} = 1+(-5)(px)+\frac{(-5)(-6)(px)^2}{2!}+\frac{(-5)(-6)(-7)(px)^3}{3!}$	

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Answer 5 of 6 · Question 7 part b · 4 marks

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7(a)	$\left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}} = 0 \Rightarrow \cos 3\theta = -\sqrt{3}\sin 3\theta \Rightarrow \tan 3\theta = -\frac{1}{\sqrt{3}}$	M1
	$\Rightarrow 3\theta = -\frac{\pi}{6}$ **or** $\frac{5\pi}{6}$	A1
	$\Rightarrow m = -\frac{\pi}{18}$ $n = \frac{5\pi}{18}$	A1 [3]
(b)	$V = \pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$	M1
	$V = \pi\left[\frac{\sin 3\theta}{3} - \frac{\sqrt{3}\cos 3\theta}{3}\right]_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}$	M1
	$V = \pi\left[\left(\frac{\sin 3\left(\frac{5\pi}{18}\right)}{3} - \frac{\sqrt{3}\cos 3\left(\frac{5\pi}{18}\right)}{3}\right) - \left(\frac{\sin 3\left(-\frac{\pi}{18}\right)}{3} - \frac{\sqrt{3}\cos 3\left(-\frac{\pi}{18}\right)}{3}\right)\right]$	M1
	$V = \frac{4\pi}{3}$	A1 [4]
		Total 7 marks

Part	Mark	Notes
(a)	M1	Sets the equation = 0 and obtains $\tan 3\theta = k$
	A1	Finds at least one correct value of 3θ $3\theta = -\frac{\pi}{6}$ **or** $\frac{5\pi}{6}$ Works in degrees Accept -30° **or** 150° for this mark. This mark is also implied by one correct solution for m or n in degrees or radians. **NB This is an M mark in Epen**
	A1	For $m = -\frac{\pi}{18}$ $n = \frac{5\pi}{18}$ which must be in radians. Accept embedded in coordinates. M and n do not need to be identified.

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(b)	**Working in degrees**. Allow working in degrees up to the last M mark.	
	M1	For a correct statement for the volume of revolution with π and their limits. Allow $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\left[\cos 3\theta+\sqrt{3}\sin 3\theta\right]^{\frac{1}{2}}\right)^2 \mathrm{d}\theta$ or $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$ Allow also: $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}x$ or even $V=\pi\int_{-\frac{\pi}{18}}^{\frac{5\pi}{18}}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)$ **This mark can be implied by correct further working** Working in degrees $V=\pi\int_{-10^\circ}^{50^\circ}\left(\cos 3\theta+\sqrt{3}\sin 3\theta\right)\mathrm{d}\theta$
	M1	For an acceptable attempt at integration. Minimally acceptable integration is as follows. $\cos 3\theta \Rightarrow \pm\frac{\sin 3\theta}{3}$, $\sin 3\theta \Rightarrow \pm\frac{\cos 3\theta}{3}$ Ignore absence or incorrect limits and the absence of π for this mark.
	M1	For substitution of the correct limits into their integrated expression the correct way around. This must be a changed expression from the one given. If the integrated expression is correct with correct limits, allow a final volume of $\frac{4\pi}{3}$ seen without explicit substitution. If the final volume is incorrect without evidence of explicit substitution award M0. If the integrated expression is incorrect or the limits are incorrect, explicit substitution **must** be seen for the award of this mark. **Do not** allow use of degrees at this stage. Ignore absence of π for this mark.
	A1	For the correct volume $(V)=\frac{4\pi}{3}$

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Question	Scheme	Marks

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number		
11 (a)(i)	$(a=)-8$	B1 [1]
(a)(ii)	$e^{2x}=9\Rightarrow 2x=\ln 9$	M1
	$x=\frac{1}{2}\ln 9=\ln 3$ *	A1cso [2]
ALT	$e^{2x}=9\Rightarrow\left[\left(e^{x}\right)^{2}=3^{2}\right]\Rightarrow e^{x}=3$	M1
	$x=\ln 3$ *	A1cso
(b)	$\pi\int_{0}^{\ln 3}\left(e^{2x}-9\right)^{2}dx$	M1
	$\pi\int_{0}^{\ln 3}\left(e^{4x}-18e^{2x}+81\right)dx$	A1
	$(\pi)\left[\frac{1}{4}e^{4x}-9e^{2x}+81x\right]_{(0)}^{("\ln 3")}$ oe	M1 A1ft
	$(\pi)\left[\left(\frac{81}{4}-81+81\ln 3\right)-\left(\frac{1}{4}-9\right)\right]$	M1
	$\pi(81\ln 3-52)$	A1 [6]
		Total 9 marks

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π must only be present for the final A1 mark and the first M1 mark (can be implied by first or final A1)

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